

SHIVAM SINGH

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SUMMARY

AI Engineer with **3+ years** of experience building **production-grade Generative AI** systems and scalable **backend services** using **Node.js**, **Python**, and **AWS**. Expert in **Large Language Models (LLMs)**, **Retrieval-Augmented Generation (RAG)**, **vector databases**, **embeddings**, and **multimodal AI** for enterprise use cases. Experienced in **agentic AI workflows**, **prompt engineering**, **Vision-Language Models (VLMs)**, and cloud-native deployments using **Docker** and **Kubernetes**.

EXPERIENCE

Associate Software Engineer

FEB 2023 – PRESENT

Raytheon Technologies, Bangalore, India

- **Inbound Material Digitization System** (Node.js, Express.js, AWS, SQL, **Vision-Language Model – Qwen2B**):
 - Designed and deployed a scalable **document digitization backend** to process inbound raw-material documents from multiple vendors, eliminating manual data entry and saving **100+ hours/month**.
 - Integrated a **Vision-Language Model (Qwen2B VLM)** for intelligent document understanding, achieving **40% improvement in extraction accuracy** over traditional OCR handling tables, handwritten fields, and multi-format invoices.
 - Engineered an end-to-end **GenAI data extraction pipeline** with **prompt engineering**, image normalization, structured JSON parsing, and post-processing validation, reducing manual correction effort by **60%**.
 - Processed 10,000+ documents/month across 20+ vendors **Used by 300+ internal employees**.
 - Built secure **RESTful APIs** using **Node.js** and **Express.js** with **JWT-based authentication** and **Single Sign-On (SSO)**; enabled centralized **SQL-based storage** for analytics and reporting.
 - Automated end-to-end document workflows, eliminating paper-based ledger systems and reducing supply chain intake processing time by **70%**.
- **HR Assistant — Generative AI Chatbot (LLaMA 3.1-70B, LangChain, RAG, ChromaDB, Streamlit, OCR)**:
 - Built an **enterprise-grade AI chatbot** powered by **LLaMA 3.1-70B** for natural language querying of internal HR policy documents, reducing average employee query resolution time by **80%**.
 - Implemented a **multimodal document ingestion pipeline** with **text embeddings** and **vector storage (ChromaDB)** supporting PDFs and scanned images via **OCR**, enabling rich semantic retrieval across heterogeneous document formats.
 - Optimized **Retrieval-Augmented Generation (RAG)** pipeline using **sentence-window retrieval** via **LangChain**, achieving **35% improvement in response relevance** and significantly reducing hallucinations.
 - Leveraged **LangGraph** for stateful **multi-turn conversation management**, enabling complex reasoning chains and persistent context across extended user sessions.
- **Internal Workflow Automation** (Node.js, Express.js, SQL, AWS, REST APIs):
 - Designed and deployed **Node.js backend microservices** to integrate production-line testing devices with centralized enterprise systems, replacing entirely manual paper-based processes.
 - Built **real-time data ingestion pipelines** to capture, process, and persist device telemetry into a **SQL database**, saving **3 hours/day** and eliminating **500+ pages** of daily paperwork.
 - Improved **system observability** through centralized logging and structured error reporting, reducing incident response time by **50%** for the operations team.

SKILLS

Generative AI and LLMs: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Multimodal AI, Vision-Language Models (VLMs), Agentic AI, Fine-Tuning, Embeddings, Natural Language Processing (NLP)
Vector Databases and Search: ChromaDB, FAISS, Pinecone, Weaviate, Vector Search, Semantic Search
AI Frameworks: LangChain, LangGraph, PyTorch, TensorFlow, Scikit-learn, Hugging Face
Languages: Python, JavaScript, TypeScript, SQL, C, C++
Frontend and Backend: Node.js, Express.js, FastAPI, Flask, REST APIs, React.js, Redux, HTML, CSS
Databases: MySQL, MongoDB, Redis
Cloud and DevOps: AWS, Docker, Kubernetes, Terraform, GitHub Actions, CI/CD Pipelines
Tools: Git, GitHub, Postman, VS Code, Agile/Scrum

EDUCATION

Master of Computer Applications (MCA)

National Institute of Technology, Tiruchirappalli (NIT Trichy)

July 2020 – June 2023

GPA: 8.01 / 10.0